

RinneFormer - Transformer-based Real-world Cooperative Perception Algorithm

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Abstract—Cooperative perception enables connected autonomous vehicles to overcome occlusions and limited sensing range by sharing complementary viewpoints. Existing feature-level fusion methods, however, often struggle to preserve fine-grained geometric structure within pillars, maintain spatial continuity across pillars, and remain robust to timing misalignment in realistic multi-agent settings. We introduce RinneFormer, a cooperative 3D object detection framework that addresses these challenges through two components: (1) a Point Transformer Feature Extractor (PTFE) that applies localized self-attention within dense pillars to capture intra-pillar geometry, and (2) a Localized Context Awareness (LCA) module that aggregates features across neighboring pillars to improve spatial coherence.

RinneFormer achieves state-of-the-art accuracy on V2XSet and strong generalization to the real-world DAIR-V2X dataset. To assess deployability, we implement a distributed two-agent pipeline in ROS, where independent RinneFormer nodes exchange intermediate features and operate under 0–100 ms of injected jitter. The system maintains stable end-to-end latency (mean: 0.32 s) and consistent detection intervals over more than 8,000 replayed frames; it also processes static LiDAR data from our EcoPRT platform, confirming compatibility with real sensor characteristics. These results demonstrate both the algorithmic effectiveness and practical system robustness of RinneFormer for cooperative perception in connected autonomous vehicles. The official implementation of the RinneFormer is available at <https://github.com/DeveshwarH1996/RinneFormer.git>.

I. INTRODUCTION

Connected autonomous vehicles (CAVs) can improve road safety by extending perception beyond the line of sight of a single platform. While modern LiDAR-based detectors such as TransFusion [1], FSD [2], and VoxelNeXt [3] achieve strong single-vehicle performance, they remain limited by occlusion, range sparsity, and blind intersections. Cooperative perception, where vehicles and infrastructure nodes share complementary viewpoints, has therefore become essential for reliable situational awareness in complex environments (Fig. 1).

Despite substantial progress, several limitations hinder practical multi-agent deployment. **Fidelity**: pillar-based encoders struggle to capture fine-grained geometry in dense regions. **Continuity**: voxelization fragments large objects, yet many fusion backbones process pillars independently, yielding inconsistent spatial features. **Robustness**: real V2X

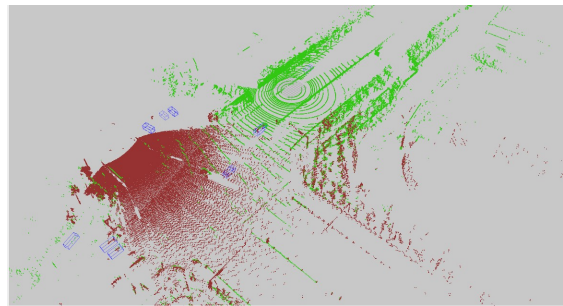


Fig. 1. Intersection scenario with box proposals from DAIR-V2X implemented in ROS-RViz2. The ego vehicle (green point cloud) observes regions beyond its sensor range by leveraging shared V2X perception (infrastructure point cloud in brown). The blue boxes represent the object proposals on the ego vehicle generated by RinneFormer.

communication introduces jitter, delay, and asynchronous capture, whereas most benchmarks assume perfect synchronization. **Deployability**: systems must balance accuracy, bandwidth, and runtime for embedded or distributed platforms.

These challenges motivate **RinneFormer**¹, a transformer-based cooperative 3D detector designed to enhance geometric fidelity, spatial continuity, and resilience to temporal misalignment. It introduces two complementary modules: (i) **Point Transformer Feature Extractor (PTFE)**, which applies localized self-attention within dense pillars to capture fine-grained geometry beyond PointNet-style encoders; and (ii) **Localized Context Awareness (LCA)**, a structured cross-attention mechanism that aggregates neighborhood context in the pseudo-image to restore continuity across fragmented pillars.

To bridge the gap between offline evaluation and deployable systems, we implement a distributed multi-agent pipeline in ROS². LiDAR sweeps from V2XSet [4], DAIR-V2X [5], and static scans from the EcoPRT vehicles [6], [7] are replayed as timestamped PointCloud2 streams for two agents. Each agent runs an independent RinneFormer node on dedicated GPUs, exchanging intermediate feature maps over ROS. We inject randomized publication delays to emulate real V2X behavior, enabling evaluation under timing jitter in a live distributed execution environment.

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¹The name RinneFormer is inspired by the concept of the Rinnegan from the anime Naruto, specifically its ability to share visual information.

²The ROS implementation of RinneFormer is available at <https://github.com/DeveshwarH1996/ros2-object-detection-cooperative.git>.

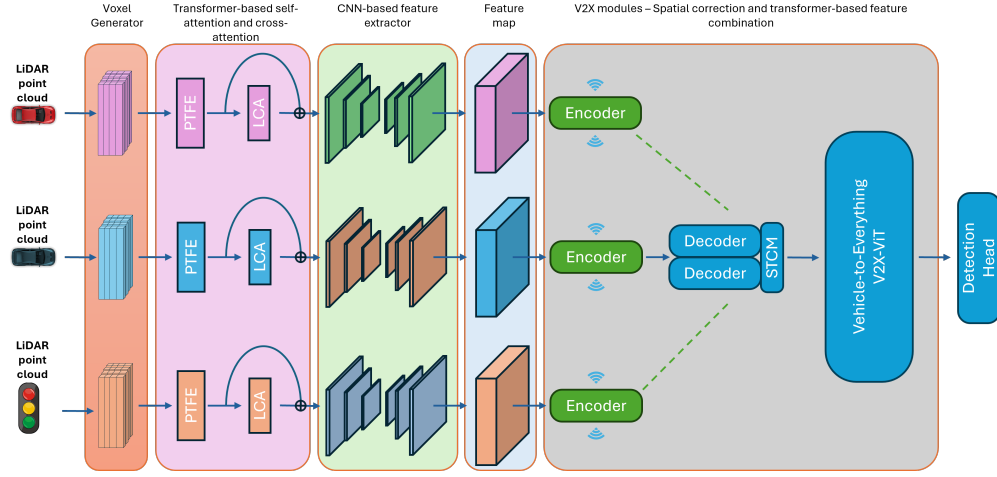


Fig. 2. Architecture of RinneFormer. LiDAR points are pillarized and processed using the proposed PTFE and LCA modules, then fused via a V2X-ViT-style encoder to produce 3D object predictions.

RinneFormer achieves state-of-the-art results on V2XSet, strong generalization to DAIR-V2X, improved robustness under noise and asynchrony, and sustained operation in a distributed multi-agent ROS deployment.

Contributions:

- 1) We propose PTFE and LCA, two transformer-based modules that improve intra-pillar geometric fidelity and inter-pillar spatial continuity.
- 2) We perform extensive evaluations on V2XSet and DAIR-V2X, including robustness experiments under noise and temporal misalignment.
- 3) We develop a distributed ROS-based cooperative perception system with real LiDAR replay, multi-GPU execution, and injected timing jitter, demonstrating practical multi-agent deployability.

II. RELATED WORK

A. Single-Vehicle 3D Object Detection

Modern LiDAR-based 3D object detectors such as TransFusion [1], FSD [2], and VoxelNeXt [3] have achieved strong performance using transformer-based fusion, sparse convolutions, and unified BEV encoders. Nonetheless, long-range sparsity, occlusion, and blind intersections fundamentally limit single-vehicle perception and motivate cooperative approaches.

B. Cooperative Perception and Fusion Strategies

Cooperative perception frameworks are typically categorized by their fusion strategy. *Early fusion* aggregates raw point clouds from multiple agents, achieving high accuracy but requiring large bandwidth [8], [9]. *Late fusion* exchanges object proposals [5], providing low communication cost but suffering under occlusion due to limited recall. *Intermediate fusion* shares learned feature maps and offers a practical balance between bandwidth and perceptual richness, becoming the dominant strategy in recent V2X LiDAR systems.

C. Transformer-Based Cooperative Perception

Transformers have been widely adopted for feature-level fusion. V2X-ViT [4] introduced transformer-based cross-agent integration with delay-aware positional encoding. CoBEVT [10] extended this idea to BEV segmentation with sparse transformers; CoMamba [11] leveraged state-space models for efficient attention; and CoopDETR [12] and Where2Comm [13] explored communication-efficient fusion.

However, most methods still rely on PointNet-style pillar encoders that treat pillars as unordered point sets, limiting the modeling of density-dependent geometric structure. Likewise, transformer backbones generally operate on pseudo-images with weak spatial continuity modeling, which becomes important when objects extend across multiple grid cells.

D. Temporal Misalignment and Delay Compensation

V2X systems must contend with real-world temporal asynchrony arising from network delay, sensor misalignment, and computation jitter. Datasets such as DAIR-V2X [5] document communication delays, while methods like CoBEVFlow [14] use motion-based warping to mitigate temporal drift. Yet most evaluations rely on offline or idealized simulations, neglecting the complex timing behaviors of live distributed systems (e.g., message arrival order, ROS callback latency, and runtime jitter).

E. System-Level Cooperative Perception

Most cooperative perception literature evaluates models offline with synchronized datasets or simulation replay. Only limited prior work studies multi-agent deployment using robotics middleware, and such efforts focus primarily on communication efficiency rather than perception robustness. To date, systematic evaluation of transformer-based cooperative perception under realistic multi-agent execution—featuring distributed nodes exchanging feature maps with non-deterministic timing—remains sparse.

F. Summary

Prior work has advanced cooperative perception in fusion strategies, transformer design, and delay modeling, but three gaps remain: (1) insufficient modeling of density-dependent geometry within pillars, (2) limited continuity modeling across neighboring pillars, and (3) lack of system-level evaluation under practical temporal misalignment. Our work addresses these challenges through a density-aware Point Transformer Feature Extractor (PTFE), a Localized Context Awareness (LCA) mechanism for spatial continuity, and a ROS-based distributed evaluation pipeline that injects realistic timing jitter.

III. METHODOLOGY

RinneFormer builds on the V2X-ViT [4] intermediate-fusion framework, introducing two new modules—Point Transformer Feature Extractor (PTFE) and Localized Context Awareness (LCA)—designed to improve geometric fidelity and spatial continuity in cooperative LiDAR perception. An overview of the architecture is shown in Fig. 2.

A. Architecture Overview

Following the V2X-ViT pipeline, LiDAR point clouds are first voxelized into pillars [15] to form a sparse pseudo-image. RinneFormer replaces the original PointNet-based pillar encoder with PTFE and augments the resulting pseudo-image with LCA before passing features to a VGG-style CNN backbone. The processed features are compressed and broadcast to neighboring agents, then decompressed, aligned using the Spatial-Temporal Correction Module (STCM), and fused through HMSA and MSwin attention layers as in [4]. The fused BEV features are finally decoded with an SSD-style detection head [16].

a) Design Rationale.: PTFE and LCA are designed specifically for cooperative LiDAR perception rather than as generic transformer modules. PTFE is density-aware: localized self-attention is applied only to high-density pillars, while sparse pillars use lightweight PointNet features, improving geometry without significant compute. LCA serves as an inductive bias that repairs local structure after pillarization and before cross-agent fusion: it performs structured, small-kernel cross-attention across neighboring pillars to explicitly restore spatial continuity lost during pillarization. This local and low-cost operation is complementary to the downstream HMSA and MSwin fusion modules, which focus on cross-agent integration and larger-range feature modeling. Together, these modules target failure modes unique to multi-agent fusion: heterogeneous point densities, fragmented object structure, and misaligned cross-agent geometry.

B. Point Transformer Feature Extractor (PTFE)

LiDAR point clouds exhibit strong spatial density variation. Traditional pillar encoders treat each pillar as an unordered point set, limiting their ability to capture fine-grained geometry in dense regions. PTFE applies transformer-based self-attention within dense pillars Fig. 3.

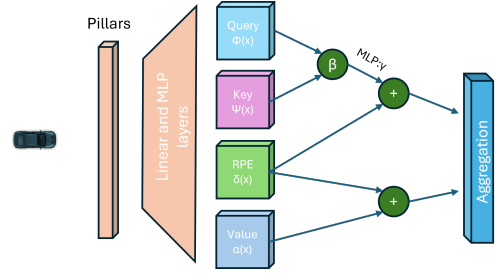


Fig. 3. Point Transformer Feature Extractor (PTFE) layer applied to a single pillar.

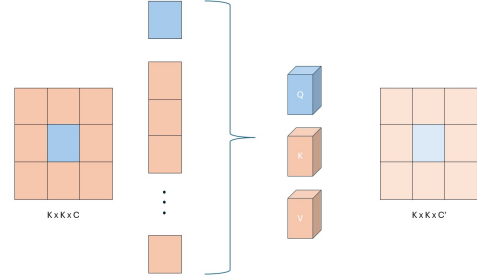


Fig. 4. Localized Context Awareness (LCA). For each $k \times k$ neighborhood ($k \times k \times C$), LCA computes cross-attended features ($k \times k \times C'$), which are concatenated with the original features.

Given points $\{x_i\}$ within a pillar, PTFE applies a Point Transformer block to model local geometric relationships:

$$y_i = \sum_{x_j \in \mathcal{N}(i)} \rho(\gamma(\phi(x_i) - \psi(x_j) + \delta)) (\alpha(x_j) + \delta), \quad (1)$$

where ϕ, ψ, α are pointwise mappings, γ is an MLP, ρ is normalization, and δ is a learnable positional encoding. Only high-density pillars (i.e., those with sufficient points, experimentally determined to be 50 points) use this attention-based encoding; low-density pillars fall back to the lightweight PointNet-style encoder from V2X-ViT. The resulting vectors are scattered back to their grid locations to form an enhanced pseudo-image.

To further improve geometric expressiveness, we optionally employ a multi-head variant following standard practice, allowing each head to model different local relationships without additional structural changes.

C. Localized Context Awareness (LCA)

While PTFE enriches intra-pillar representations, voxelization still fragments large objects across adjacent pillars, and conventional backbones process each pillar independently. LCA addresses this by introducing structured cross-attention over local neighborhoods in the pseudo-image, as shown in Fig. 4.

For each pillar location (i, j) , LCA extracts a $k \times k$ spatial patch and applies local cross-attention:

$$\text{LCA}(i, j) = \text{softmax}(QK^\top)V, \quad (2)$$

where queries, keys, and values are computed from the patch. A small learnable positional encoding is added to preserve

spatial layout. The resulting context features are concatenated with the original pseudo-image channels, restoring spatial continuity across fragmented object surfaces and improving downstream multi-agent fusion. We use $k=3$ in all experiments.

D. Model Complexity

TABLE I
MODEL COMPLEXITY.

Model	Params (M)	Latency (ms)
V2X-ViT	13.4	140
RinneFormer	13.5	154

RinneFormer remains lightweight, increasing V2X-ViT's 13.4M parameters to only 13.5M and offline latency from 140 ms to 154 ms, a small overhead relative to its accuracy gains.

E. Integration with Cooperative Fusion

After PTFE and LCA processing, the augmented pseudo-image is forwarded through a VGG-style CNN backbone, compressed via a 1×1 convolution, and transmitted to neighboring agents. STCM is applied after feature reception and before cross-agent fusion, following the V2X-ViT formulation without modification [17]. The received features are then fused through HMSA and MSwin modules from V2X-ViT. An SSD-style head produces final 3D bounding box predictions.

This integration enables PTFE and LCA to enhance both local geometric fidelity and inter-pillar spatial coherence while remaining fully compatible with existing V2X intermediate-fusion pipelines.

IV. EVALUATION

A. Experimental Setup

a) Training.: All models are trained using the Adam optimizer with an initial learning rate of 10^{-3} and a cosine annealing schedule with warm restarts. The warm-up learning rate is set to 1.5×10^{-3} for the first five epochs. Training and offline evaluation are conducted using the OpenCOOD framework [18] on an NVIDIA RTX 3090 Ti GPU.

b) ROS Multi-Agent Execution.: To examine deployability, we implement RinneFormer in a distributed two-agent ROS architecture (Fig. 5). A replay node publishes DAIR-V2X [5] or V2XSet [4] LiDAR sweeps as timestamped PointCloud2 messages, along with pose and velocity information. Two RinneFormer nodes, each running on a dedicated GPU (dual RTX 3090 Tis), process their respective LiDAR streams, exchange intermediate features via ROS topics, and perform feature-level fusion.

c) Temporal Jitter and Real Sensor Data.: To model realistic V2X timing irregularities, we inject independent per-frame publication delays, sampled uniformly from 0–100 ms. Additionally, static EcoPRT vehicle LiDAR bag files [6], [7] containing point clouds and GPS data are replayed to verify compatibility with real sensor noise; although unlabeled, these support latency measurement and execution-stability assessment.

B. Results on V2XSet

Table II shows that RinneFormer achieves state-of-the-art performance on V2XSet, improving by +1.3 AP@0.5 and +4.0 AP@0.7 over V2X-ViT, reflecting gains in both detection recall and localization accuracy.

TABLE II
PERFORMANCE COMPARISON ON V2XSET DATASET.

Method	Year	AP@0.5	AP@0.7
F-Cooper [8]	2019	84.0	68.0
V2VNet [19]	2020	84.5	67.7
Attentive Fusion [18]	2021	80.7	66.4
DiscoNet [20]	2022	84.4	69.5
CoBEVT [10]	2022	84.9	66.0
Where2Comm [13]	2022	85.5	65.4
V2X-ViT [4]	2022	88.2	71.2
CoAlign [21]	2023	85.8	76.5
CoMamba [11]	2024	88.3	72.9
CoopDETR [12]	2025	86.9	76.5
RinneFormer	-	89.5	80.5

RinneFormer outperforms the strongest prior baseline, CoopDETR, by +4.0 AP@0.7, demonstrating improved localization stability under the same intermediate-fusion setting.

C. Results on DAIR-V2X

RinneFormer generalizes strongly to the real-world DAIR-V2X dataset (Table III), outperforming V2X-ViT by +7 AP@0.5 and +2 AP@0.7 without dataset-specific tuning.

TABLE III
PERFORMANCE ON DAIR-V2X.

Method	AP@0.5	AP@0.7
V2X-ViT [4]	52.0	32.0
RinneFormer Base	55.0	33.0
RinneFormer MH+LCA	59.0	34.0

D. Robustness to Noise and Temporal Misalignment

We evaluate noise and synthetic delay robustness using OpenCDA on V2XSet. RinneFormer maintains higher performance than V2X-ViT under injected noise and asynchrony (Table IV).

a) Runtime Jitter in ROS.: Under 0–100 ms randomly injected delays in both agents' LiDAR streams, the distributed ROS pipeline processed more than 8,000 replayed frames without node crashes or ROS queue overflows. The inter-detection interval remained tightly concentrated around

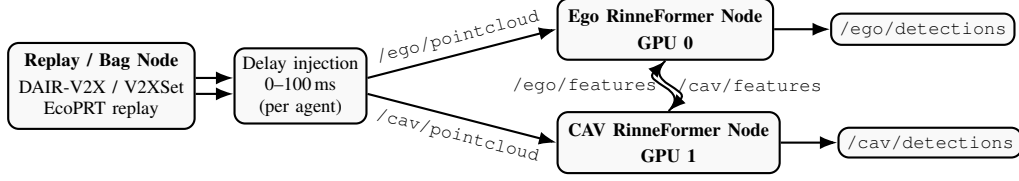


Fig. 5. ROS-based multi-agent cooperative perception architecture. A replay/bag node publishes LiDAR (PointCloud2), pose, and velocity messages from DAIR-V2X/V2XSet and EcoPRT recordings. Independent 0–100 ms delays are injected into each agent’s LiDAR stream. Two RinneFormer nodes run on separate GPUs, subscribe to their respective point clouds, exchange intermediate features via ROS topics, and produce ego- and CAV-frame detections.

TABLE IV
ROBUSTNESS UNDER OPENCDA-INJECTED NOISE/ASYNCHRONY.

Model	AP@0.5	AP@0.7
V2X-ViT [4]	84.0	61.0
RinneFormer MH+LCA	86.0	68.0

0.5 s, indicating stable behavior under runtime misalignment, even though not every LiDAR frame results in an emitted detection.

E. Distributed ROS Runtime and Bandwidth

Each agent achieves a mean end-to-end inference latency of 320 ± 83 ms on RTX 3090 Ti GPUs. Intermediate feature maps require approximately 27 Mbps bandwidth, similar to V2X-ViT. The PyTorch model does not yet meet real-time rates on embedded hardware (Jetson AGX Orin), and TensorRT optimization is required for on-board deployment.

TABLE V
LATENCY MEASUREMENTS FOR EACH STAGE OF THE RINNEFORMER
ROS EXECUTION PIPELINE

Metric Overhead Type	CAV 0 (s)	CAV 1 (s)
ROS Publisher	0.008 ± 0.074 s	0.004 ± 0.044 s
Receive → Encoder Start	0.001 ± 0.000 s	0.001 ± 0.000 s
Encoder Duration	0.094 ± 0.055 s	0.050 ± 0.051 s
Encoder→Fusion start	0.160 ± 0.003 s	0.160 ± 0.002 s
Fusion Duration	0.073 ± 0.005 s	0.072 ± 0.004 s
Finish → Detection Pub.	0.011 ± 0.003 s	0.013 ± 0.004 s
Publish → Detection (E2E)	0.342 ± 0.095 s	0.298 ± 0.062 s
Global Metrics:		
Inter-detection interval: 0.497 ± 0.029 s		
True E2E Latency: 0.320 ± 0.083 s		

a) *End-to-End Latency Analysis.*: We further profile timing behavior across the full ROS-based multi-agent pipeline on DAIR-V2X replay with 0–100 ms injected jitter at the time of transmitting intermediate features. Table V reports per-stage latency for both agents. ROS message handling is lightweight, with publish-to-receive delays below 10 ms on average. The encoder stage accounts for 50–94 ms per frame. The encoder-time difference between CAV 0 and CAV 1 is due to viewpoint-dependent pillar density: depending on the scene, one agent observes denser regions than the other, increasing PTFE computation in high-density pillars. This variation is expected in cooperative LiDAR setups and does not affect fusion, which waits for both feature sets.

The ~ 160 ms delay between encoder output and fusion start is a system-level effect. After encoding, features are copied to the CPU, reshaped and serialized for ROS transmission; the receiving agent performs deserialization and CPU–GPU transfer before concatenating remote features with its own. This processing, combined with asynchronous arrival under 0–100 ms jitter, accounts for the pre-fusion wait time, after which fusion executes immediately. The fusion module itself is stable across agents (0.073 ± 0.005 s for CAV 0 and 0.072 ± 0.004 s for CAV 1). End-to-end perception latency, measured from LiDAR publish to detection output, averages 0.342 ± 0.095 s for CAV 0 and 0.298 ± 0.062 s for CAV 1, with a global mean of 0.320 ± 0.083 s. The inter-detection interval remains stable at 0.497 ± 0.029 s, indicating consistent system behavior under 0–100 ms jitter, although not every LiDAR frame yields a separate detection given the current model latency and asynchronous execution.



Fig. 6. EcoPRT vehicle setup for data capture

F. EcoPRT Static-Data Validation

Replaying static EcoPRT LiDAR bag files (Fig. 6) through the ROS pipeline verifies that RinneFormer handles real sensor noise, calibration offsets, and timing characteristics despite the absence of ground-truth labels.

G. Ablation Study

Ablation results on V2XSet (Table VI) quantify the contributions of the proposed PTFE (multi-head attention) and

TABLE VI
ABLATION STUDY ON THE IMPACT OF PTFE AND LCA LAYERS ON
MEAN AVERAGE PRECISION.

Model	PTFE	LCA	AP@0.5	AP@0.7
Control			88.2	71.2
RinneFormer	✓	✓	88.4	78.7
	✓		89.5	80.5

LCA modules relative to the V2X-ViT baseline. Incorporating PTFE yields a substantial improvement at the stricter AP@0.7 threshold, reaching 78.7%, indicating more precise localization and higher-confidence bounding box predictions. This improvement suggests that multi-head point-level attention enables more robust aggregation of local geometric features within dense pillars.

Adding the LCA layer further improves performance across both thresholds, achieving 89.5% AP@0.5 and 80.5% AP@0.7. By enforcing structured interactions among neighboring pillars, LCA enhances spatial consistency in the fused feature map, leading to more coherent detections and complementary gains when combined with PTFE.

LIMITATIONS

Due to latency and asynchronous execution, not all LiDAR frames produce detections. Experiments are conducted on a dual-GPU workstation, with real-time deployment requiring TensorRT optimization. EcoPRT data are static and unlabeled, and communication effects beyond 0–100 ms jitter are not modeled. Despite two-agent evaluation, RinneFormer naturally supports N -agent fusion.

V. CONCLUSION

We presented RinneFormer, a cooperative 3D perception framework that improves geometric fidelity and spatial continuity via PTFE and LCA. RinneFormer achieves state-of-the-art performance on V2XSet and strong generalization to DAIR-V2X, and is validated in a distributed two-agent ROS pipeline operating robustly under 0–100 ms temporal jitter. Latency profiling and EcoPRT LiDAR replay confirm its system-level viability.

FUTURE WORK

Future work will target real-time embedded deployment via TensorRT, mixed-precision inference, and model compression, alongside collecting fully annotated dynamic multi-vehicle LiDAR data on the EcoPRT platform. We also plan to extend RinneFormer to larger, heterogeneous multi-agent networks and richer communication effects.

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